Spaced Retrieval Practice in Introductory Programming: A Small Classroom Study

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Abstract

We test whether spacing short retrieval quizzes over several days helps novice programmers remember loop and function concepts better than massing the same quizzes into one sitting. In a classroom study with 48 students, the spaced group gained an average of 14 points on a delayed test, while the massed group gained 6 points. The benefit was largest for questions about nested loops. We conclude that low-stakes spaced quizzes are a cheap and effective addition to a first programming course.

1 Introduction

Novice programmers often feel that they understand a concept right after seeing it, and then struggle to use it a week later. Retrieval practice, the act of recalling information from memory instead of rereading it, is known to improve long-term retention. Spacing the practice over time is known to improve it further. Few studies have tested both ideas together in a real introductory programming classroom.

Our research question is simple: does spacing retrieval quizzes over days produce better delayed-test performance than taking the same number of quizzes in a single sitting? We also ask which kinds of questions benefit most.

2 Related Work

Earlier work in cognitive psychology shows the spacing effect for vocabulary and facts. Studies in programming education have mostly looked at tools, such as automated feedback, rather than at the timing of practice. Our study is closest to classroom experiments that vary the schedule of homework, but we use short quizzes that take under five minutes.

3 Method

3.1 Participants

Forty-eight students in two sections of an introductory programming course took part. Students were assigned to the spaced group or the massed group by section, and neither group had prior programming experience beyond one week of the course.

3.2 Procedure

Both groups took four five-question quizzes on loops and functions. The spaced group took one quiz every three days, so the quizzes covered twelve days. The massed group took all four quizzes back to back on the first day. Each quiz asked students to write short answers from memory, and correct answers were shown after each quiz. Twelve days after the last quiz of the spaced group, every student took a 20-question delayed test.

3.3 Measures

The main measure was the gain from a pre-test to the delayed test, in percentage points. We also scored the nested-loop questions separately, because instructors reported that these were the hardest for beginners.

4 Results

The spaced group gained 14 points on average from the pre-test to the delayed test, compared with 6 points for the massed group. The difference was statistically significant. On the nested-loop questions the spaced group’s advantage grew to 19 points over the massed group.

Students in the spaced group also reported feeling less confident after each quiz than the massed group did, even though they remembered more later.

5 Discussion

The results suggest that the timing of practice matters as much as its amount. Spacing forces students to retrieve a concept after they have partly forgotten it, and this effortful recall strengthens memory. The lower confidence reported by the spaced group is consistent with the idea of desirable difficulties: practice that feels harder can work better.

The study has limits. Students were assigned by section instead of at random, the sample is small, and we measured memory twelve days after practice rather than at the end of the semester.

6 Conclusion

Short quizzes spaced three days apart produced larger delayed-test gains than the same quizzes taken at once. Instructors can adopt this at almost no cost by spreading existing practice questions across the week. Future work should use random assignment and a longer retention interval.

References

[1] Example Author. The Spacing Effect in Memory. Journal of Imaginary Learning, 2015.

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